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When $\beta(1+r)=1$; $u(c) = \alpha c - \frac{\gamma}{2} c^2$,

$$\Delta C_{t+1} = \frac{r}{1+r} \sum_{j=0}^{\infty} \frac{1}{(1+r)^j} [E_{t+1} y_{t+1+j} - E_t y_{t+1+j}]$$

↳ When $j=0$: $E_{t+1} \dots = E_{t+1}$

↳ When $j=1$: $E_{t+1} \dots = E_{t+1} [\rho^2 y_t + \rho \epsilon_{t+1} + \epsilon_{t+2}^0]$
 $- E_t [\rho^2 y_t + \rho \epsilon_{t+1}^0 + \epsilon_{t+2}^0]$

$$= \rho \epsilon_{t+1}$$

↳ When $j=2$: $E_{t+1} \dots = E_{t+1} (\rho^3 y_t + \rho^2 \epsilon_{t+1} + \rho \epsilon_{t+2} + \epsilon_{t+3}^0)$
 $- E_t (\rho^3 y_t + \rho^2 \epsilon_{t+1} + \rho \epsilon_{t+2} + \epsilon_{t+3}^0)$

↳ $\frac{r}{1+r} \left(\epsilon_{t+1} + \rho \frac{\epsilon_{t+1}}{(1+r)} + \rho^2 \frac{\epsilon_{t+1}}{(1+r)^2} + \dots \right)$

$$\frac{r}{1+r} \epsilon_{t+1} \cdot \frac{1}{1 - \frac{\rho}{1+r}} = \frac{r}{1+r} \epsilon_{t+1} \frac{1+r}{1+r-\rho}$$

$$= \Delta C_{t+1} = \epsilon_{t+1} \frac{r}{1+r-\rho}$$

Consider:

$$\textcircled{1} \rho = 1 \Rightarrow y_t = y_{t-1} + \varepsilon_t$$

$$y_{t+1} = y_{t-1} + \varepsilon_t + \varepsilon_{t+1}$$

$$y_{t+2} = y_{t-1} + \varepsilon_t + \varepsilon_{t+1} + \varepsilon_{t+2}$$

Impact of shock never declines,
always persists

$$\Delta c_{t+1} = \varepsilon_{t+1} \cdot \frac{r}{1+r-1} = \varepsilon_{t+1} \cdot 1 = \varepsilon_{t+1}$$

$$\textcircled{2} \rho = 0 \Rightarrow y \text{ is i.i.d.}$$

$$\Delta c_{t+1} = \frac{r}{1+r} \varepsilon_{t+1}$$

How to find Consumption Function:

\textcircled{1} Brute Force:

$$c_t = r a_t + \frac{r}{1+r} E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j}$$

$$y_t = \rho y_{t-1} + \varepsilon_t$$

↳ substitute into $c_t = \dots$

$$E_t y_t + \frac{E_t (\rho y_t + \varepsilon_{t+1})}{(1+r)} + \frac{E_t (\rho^2 y_t + \rho \varepsilon_{t+1} + \varepsilon_{t+2})}{(1+r)}$$

$$\Rightarrow r a_t + \frac{r}{(1+r)} y_t \left(\frac{1}{1 - \frac{\rho}{1+r}} \right) = r a_t + \frac{r y_t}{1+r-\rho}$$

$$c_t = r a_t + \frac{r}{1+r-\rho} y_t$$

② Guess - And - Verify

$$y_t = \rho y_{t-1} + \varepsilon_t$$

$$\text{guess: } c_t = A y_{t-1} + B \varepsilon_t + D a_t$$

$$c_{t+1} = A y_t + B \varepsilon_{t+1} + D a_{t+1}$$

$$= A \rho y_{t-1} + A \varepsilon_t + B \varepsilon_{t+1} + D a_{t+1}$$

use Euler:

$$c_t = E_t c_{t+1} \quad (\text{only bc quadratic } u(c))$$

$$A y_{t-1} + B \varepsilon_t + D a_t = E_t (A \rho y_{t-1} + A \varepsilon_t + B \varepsilon_{t+1} + D a_{t+1})$$

$$A(1-\rho)y_{t-1} + (B-A)\varepsilon_t + D a_t = D a_{t+1}$$

$$a_{t+1} = \frac{A(1-\rho)}{D} y_{t-1} + \frac{(B-A)}{D} \varepsilon_t + a_t$$

$$\text{bc: } c_t = a_t(1+r) + \rho y_{t-1} + \varepsilon_t - a_{t+1}$$

$$c_t = a_t(1+r) + \rho y_{t-1} + \varepsilon_t - \frac{A(1-\rho)}{D} y_{t-1} - \frac{(B-A)}{D} \varepsilon_t - a_t$$

$$c_t = r \cdot a_t + y_{t-1} \left(\rho - \frac{A(1-\rho)}{D} \right) + \varepsilon_t \left(1 - \frac{(B-A)}{D} \right)$$

$$D=r; A = \left(\rho - \frac{A(1-\rho)}{D} \right); B = \left(1 - \frac{(B-A)}{D} \right)$$

$$rA = \rho r = A(1-\rho) \Rightarrow \rho r = A(1-\rho-r) \Rightarrow A = \frac{\rho r}{1+r-\rho}$$

$$B = 1 - \frac{B-A}{r} \Rightarrow Br = r - B + A \Rightarrow B(1+r) = r + \frac{\rho r}{1+r-\rho}$$

$$B(1+r) = \frac{r(1+r-\rho) + \rho r}{1+r-\rho} = \frac{r(1+r) - \rho r + \rho r}{1+r-\rho}$$

$$B(1+r) = \frac{r(1+r)}{1+r-\rho} \Rightarrow B = \frac{r}{1+r-\rho}$$

$$c_t = \frac{\rho r}{1+r-\rho} y_{t-1} + \frac{r}{1+r-\rho} \varepsilon_t + r a_t$$

$$c_t = \frac{r}{1+r-\rho} (\rho y_{t-1} + \varepsilon_t) + r a_t = \frac{r}{1+r-\rho} y_t + r a_t$$

Another Income Process:

$$y_t = \underset{\substack{| \\ \text{permanent} \\ \text{income}}}{y_t^P} + \underset{\substack{| \\ \text{transitory} \\ \text{income}}}{U_t}$$

$$y_t^P = y_{t-1}^P + \mu_t \quad ; \quad U_t \sim \text{i.i.d.}$$

Brute Force:

$$E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j} = y_t^P + U_t + E_t \frac{y_t^P + \mu_{t+1} + U_{t+1}}{1+r} \\ + E_t \frac{y_t^P + \mu_{t+1} + \mu_{t+2} + U_{t+2}}{(1+r)^2} + \dots$$

$$= y_t^P + U_t + \frac{y_t^P}{(1+r)} + \frac{y_t^P}{(1+r)^2} + \dots$$

$$C_t = r a_t + \frac{r}{1+r} E_t \sum_{j=0}^{\infty} \frac{y_{t+j}}{(1+r)^j}$$

$$= r a_t + \frac{r}{1+r} \left(U_t + y_t^P \frac{1+r}{r} \right)$$

$$= r a_t + \frac{r}{1+r} U_t + y_t^P$$

↳ smooth transitory

↳ consume all permanent